Quicksort
Full Tree Trace

Note this is not how the algorithm calls recursively. It is for intuition on how it works.

For this see the quicksort recursive call trace

[4,3,2,7,9,8,6,1,5]

Divide

[4,3,2,7,9,8,6,1,5]

↓

[3,2,1] + [4] + [7,9,8,6,5]

1. Pick 4 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

1. Pick 3 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[1] + [2] + []

1. Pick 2 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[6,5] + [7] + [9,8]

[1] + [2] + []

1. Pick 7 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[6,5] + [7] + [9,8]

[1] + [2] + []

[5] + [6] + []

1. Pick 6 as the pivot
2. Place items lower to the left
3. Place items higher to the right

Divide

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[6,5] + [7] + [9,8]

[1] + [2] + []

[5] + [6] + []

[8] + [9] + []

1. Pick 9 as the pivot
2. Place items lower to the left
3. Place items higher to the right

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[6,5] + [7] + [9,8]

[1] + [2] + []

[5] + [6] + []

[8] + [9] + []

Notice that all the pivots (in red) are in sorted order from left to right!

These are then combined back together

Conquer

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[2,1] + [3] + []

[6,5] + [7] + [9,8]

[1] + [2] + []

[5] + [6] + []

[8] + [9] + []

Combine the lists

Conquer

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[1,2] + [3] + []

[6,5] + [7] + [9,8]

[5] + [6] + []

[8] + [9] + []

Combine the lists

Conquer

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[1,2] + [3] + []

[5,6] + [7] + [9,8]

[8] + [9] + []

Combine the lists

Conquer

[4,3,2,7,9,8,6,1,5]

[3,2,1] + [4] + [7,9,8,6,5]

[1,2] + [3] + []

[5,6] + [7] + [8,9]

Combine the lists

Conquer

[4,3,2,7,9,8,6,1,5]

[1,2,3] + [4] + [7,9,8,6,5]

[5,6] + [7] + [8,9]

Combine the lists

Conquer

[4,3,2,7,9,8,6,1,5]

[1,2,3] + [4] + [5,6,7,8,9]

Combine the lists

[1,2,3,4,5,6,7,8,9]

Original list is now sorted